

SHAYESHA WEERAKKODY

Assistant Research Officer (Climate Risk) | GIS & Remote Sensing Analyst

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PROFESSIONAL SUMMARY

Early-career GIS and Remote Sensing professional with a BSc (Hons) in Geographical Information Science (GIS) from the University of Peradeniya, with strong applied experience in climate risk-related geospatial analysis, irrigation and water-resource mapping, and environmental assessment. Experienced in hazard mapping (floods, droughts), remote sensing-based indicators (NDVI, NDDI, LST), GIS-based modelling, and Python-assisted spatial analysis. Demonstrated ability to integrate multi-source climate, environmental, and geospatial datasets to support climate-resilient planning, agriculture, and water-management decision-making. Highly motivated to contribute to climate risk, water, and resilience programmes.

CORE TECHNICAL COMPETENCIES

Climate & Hazard Analysis: Flood hazard mapping | Drought vulnerability assessment | Climate-smart agriculture zoning | Remote sensing-based climate indicators (NDVI, NDDI, LST, etc.) | DEM-based hydrological & watershed analysis

Geospatial & Remote Sensing Tools: ArcGIS Pro | ArcMap | ArcGIS Online | QGIS | Google Earth Engine | ERDAS IMAGINE | ENVI (basic) | AutoCAD | GeoServer | Esri CityEngine

Satellite & EO Data: Landsat | Sentinel | MODIS | Vegetation indices | Land-surface datasets | Land use / land cover classification

Programming & Data Handling: Python (GIS automation & spatial analysis) | R (basic spatial analysis) | Geodatabase management | QA/QC | Spatial data integration & processing

Visualization & Reporting: Climate & hazard maps | Analytical charts & tables | Technical reports & research documentation | ArcGIS Online dashboards & web maps

PROFESSIONAL EXPERIENCE

GIS Analyst Intern

GIS Branch, Irrigation Department – Head Office, Colombo 07

April 2025 – October 2025

- Supported GIS-based analysis for irrigation planning, flood-related assessments, and sustainable water-resource management.
 - Conducted spatial data collection, digitization, validation, and geodatabase management following institutional QA/QC standards.
 - Produced thematic, analytical, and hazard-related maps using ArcGIS Pro and QGIS to support departmental planning and reporting.
 - Assisted in remote sensing analysis for land and water-related GIS projects, integrating satellite imagery with vector-based GIS layers.
 - Collaborated with engineers, surveyors, and field officers to deliver evidence-based geospatial outputs for decision-making.
 - Contributed to technical documentation and map-based reporting for internal and stakeholder use.
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EDUCATION

Bachelor of Science (Hons) in Geographical Information Science (GIS)

University of Peradeniya | 2021 – 2025 | Cumulative GPA: 3.288

Relevant Coursework & Academic Focus:

Remote Sensing & Image Processing | GIS-based Spatial Analysis & Modelling | Hydrology & Watershed Analysis | Environmental & Agricultural GIS | Climate-related Geospatial Applications

SELECTED PROJECTS & RESEARCH

Final Year Research Project: Geospatial Delineation of Climate-Smart Agricultural Zones in Anuradhapura District: A Remote Sensing-Based Approach to Enhancing Drought Resilience.

- Developed climate-smart agricultural zoning using GIS-based Multi-Criteria Decision Analysis (MCDA).
- Conducted drought vulnerability assessment using NDVI, NDDI, and climate-related indicators.
- Performed flood hazard mapping and watershed delineation using DEM-based hydrological modelling.
- Integrated land use / land cover classification with environmental and climatic factors to support climate-resilient agricultural planning.
- Produced spatially explicit maps and analytical outputs suitable for policy and planning applications.

CERTIFICATIONS & TRAINING

Esri GIS Training Courses | Python for Beginners – University of Moratuwa | Python for Programming – University of Moratuwa | Introductory Computer Programming (Python) – University of Peradeniya | R Programming - Great Learning Academy

ADDITIONAL INFORMATION

Languages: Sinhala – Native | English – Proficient (written and spoken)

Professional Skills: Analytical & critical thinking | Climate- and data-driven problem solving | Technical reporting & documentation | Attention to detail & data accuracy | Independent & interdisciplinary team work | Strong commitment to methodological transparency & quality

REFERENCES

- Prof. T.W.M. Tilak Bandara | Head of Department, Department of Geography, University of Peradeniya | Phone: +94 77 717 9630 | Email: twmtlak@arts.pdn.ac.lk
- Mr. A.K. Wickramasooriya | Senior Lecturer, Department of Geography, University of Peradeniya | Phone: +94 77 940 6168 | Email: ashvin@arts.pdn.ac.lk
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